

IN-LAB EXERCISE:

OBJECTIVE:

To transform COVID-19 data into a structured format suitable for Business Intelligence (BI) reporting and design interactive dashboards with Key Performance Indicators (KPIs) using Tableau and Power BI.

PROCEDURE:

1. Data Collection:

- Import the **COVID-19 dataset** from inbuilt sources like *Our World in Data* or CSV files containing country-level COVID statistics (e.g., new cases, deaths, vaccinations).
- Datasets may include fields such as *date*, *location*, *new_cases*, *total_cases*, *new_deaths*, *total_deaths*, *people_vaccinated*, etc.

2. Data Cleaning and Transformation:

- Convert columns like *date* to proper date formats.
- Handle missing or null values in *new_cases*, *total_cases*, etc.
- Create calculated fields (e.g., daily case growth rate, case fatality rate, vaccination percentage).

3. Data Aggregation:

- Aggregate data by *continent*, *country*, or *month*.
- Use measures like SUM(*new_cases*), AVG(*positivity_rate*), MAX(*total_deaths*).

4. Designing KPIs:

- Identify KPIs such as:
 - Total Confirmed Cases
 - Total Deaths
 - Case Fatality Rate
 - Vaccination Coverage (%)
 - Daily Case Growth Rate
- Use DAX (Power BI) or Calculated Fields (Tableau) to compute and visualize these KPIs.

5. Dashboard Development:

- Build visualizations such as:
 - Line charts for daily new cases and deaths
 - Bar charts for country-wise vaccination progress

- Maps to show regional case distribution
- KPI cards for high-level insights

6. Publishing and Sharing:

- Publish the dashboard using Tableau Public/Server or Power BI Service.
- Share the dashboards with stakeholders through public or organizational portals.

Experiment Dataset:

COVID-19 Global Dataset

Fields Used:

- **Date** – Date of the report
- **Location** – Country or region
- **New Cases** – Daily reported cases
- **New Deaths** – Daily reported deaths
- **People Vaccinated** – Cumulative vaccinated count
- **Total Cases** – Cumulative confirmed cases
- **Total Deaths** – Cumulative deaths
- **Population** – Total population of the country

POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

- 1. How did data transformation improve the quality of insights in your COVID-19 dashboard?**
- 2. What challenges did you encounter during the data cleaning process, and how did you resolve them?**
- 3. Which KPI provided the most valuable insight, and why?**

4. How does interactivity in dashboards improve user experience?

5. What steps would you take to further enhance the dashboard's effectiveness?

RESULT:

The COVID-19 dataset was successfully cleaned, aggregated, and transformed for visual analysis using Tableau and Power BI. Missing values, data type mismatches, and inconsistencies were addressed. KPIs such as vaccination rate and daily case growth offered valuable insights into the pandemic trends across countries and timeframes. The resulting interactive dashboards allowed for effective public health monitoring and communication.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		